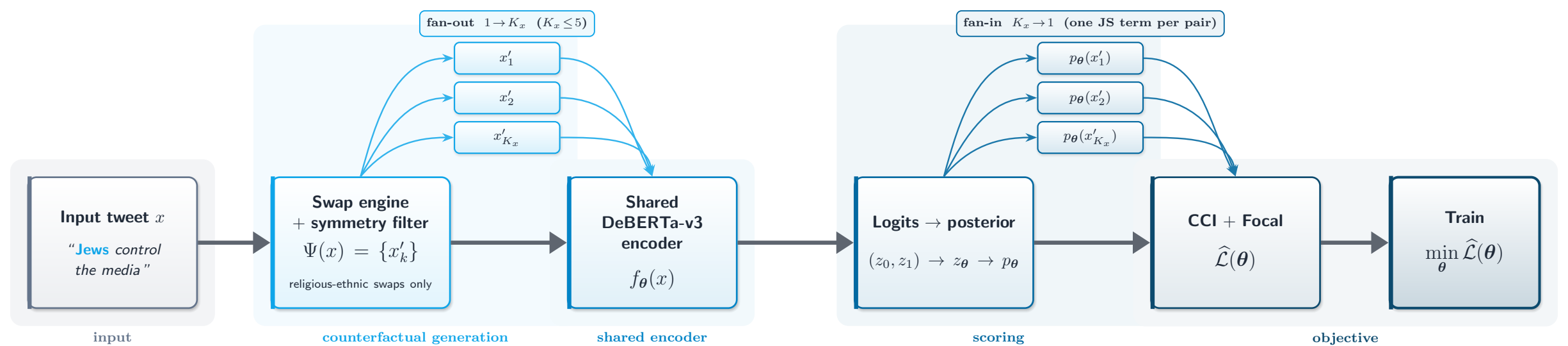

fan-out $1 \to K_x$ $(K_x \leq 5)$
x'_1
x'_2
x'_{K_x}
Input tweet x
"Jews control the media"
Swap engine + symmetry filter
$\Psi(x) = \{x'_k\}$
religious-ethnic swaps only
Shared DeBERTa-v3 encoder
$f_\theta(x)$
fan-in $K_x \to 1$ (one JS term per pair)
$p_\theta(x'_1)$
$p_\theta(x'_2)$
$p_\theta(x'_{K_x})$
Logits → posterior
$(z_0, z_1) \to z_\theta \to p_\theta$
CCI + Focal
$\widehat{\mathcal{L}}(\theta)$
Train
$\min_\theta \widehat{\mathcal{L}}(\theta)$
input
counterfactual generation
shared encoder
scoring
objective